

EXp 6: EDGE DETECTION

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```
In [18]: import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread('download (1).jpg')
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
```

Out[18]: (np.float64(-0.5), np.float64(735.5), np.float64(1103.5), np.float64(-0.5))

Original Image



```
In [19]: sobel_x = cv2.Sobel(gray_image, cv2.CV_64F, 1, 0, ksize=5)
sobel_y = cv2.Sobel(gray_image, cv2.CV_64F, 0, 1, ksize=5)
sobel_combined = cv2.magnitude(sobel_x, sobel_y)
plt.imshow(sobel_combined, cmap='gray')
plt.title('Sobel Edge Detection')
plt.axis('off')
```

Out[19]: (np.float64(-0.5), np.float64(735.5), np.float64(1103.5), np.float64(-0.5))

Sobel Edge Detection



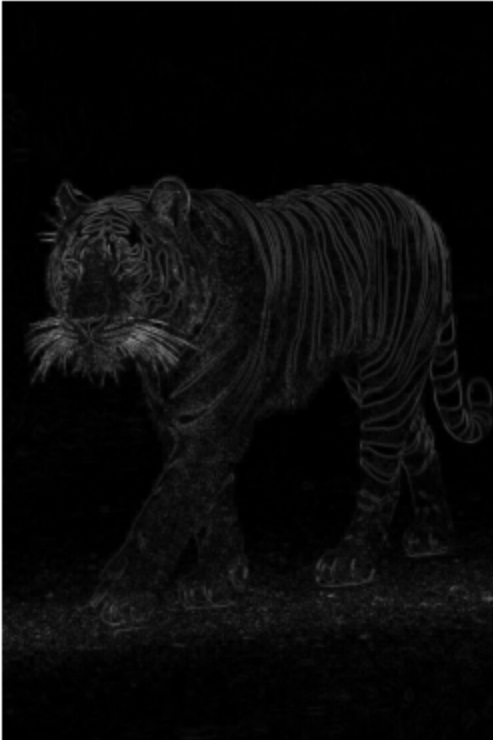
```
In [20]: prewitt_x = cv2.Sobel(gray_image, cv2.CV_64F, 1, 0, ksize=3)
prewitt_y = cv2.Sobel(gray_image, cv2.CV_64F, 0, 1, ksize=3)
prewitt = abs(prewitt_x) + abs(prewitt_y)
plt.imshow(prewitt, cmap='gray')
plt.title('Prewitt Edge Detector')
plt.axis('off')
plt.show()
```

Prewitt Edge Detector



```
In [21]: roberts_x = cv2.filter2D(gray_image, cv2.CV_64F, np.array([[1, 0], [0, -1]]))
roberts_y = cv2.filter2D(gray_image, cv2.CV_64F, np.array([[0, 1], [-1, 0]]))
roberts = abs(roberts_x) + abs(roberts_y)
plt.imshow(roberts, cmap='gray')
plt.title('Roberts Edge Detector')
plt.axis('off')
plt.show()
```

Roberts Edge Detector



```
In [22]: laplacian = cv2.Laplacian(gray_image, cv2.CV_64F)
limit = np.percentile(np.abs(laplacian), 95)
plt.imshow(laplacian, cmap='gray', vmin=-limit, vmax=limit)
plt.title('Laplacian Edge Detection')
plt.axis('off')
plt.show()
```

Laplacian Edge Detection



```
In [23]: canny_edges = cv2.Canny(gray_image, 50, 150)
plt.imshow(canny_edges, cmap='gray')
plt.title('Canny Edge Detection')
plt.axis('off')
```

Out[23]: (np.float64(-0.5), np.float64(735.5), np.float64(1103.5), np.float64(-0.5))

Canny Edge Detection

